

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Real time systems [ELE_4412]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain *dependencies among real-time tasks*. Discuss AND/OR precedence constraints and illustrate with a precedence graph example (3)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

B)

Task	Computation Time (C)	Period (T)
T ₁	2 ms	6 ms
T ₂	1 ms	8 ms
T ₃	2 ms	12 ms

(4)

Schedule these tasks using a Cyclic Scheduler with an appropriate frame size.
Show the schedule for one major cycle.

Course Outcome: "CO1" / Cognitive Level: "Application"

C)

Task	Computation Time (C)	Period (T)
T ₁	2 ms	6 ms
T ₂	1 ms	8 ms
T ₃	2 ms	12 ms

(3)

For the same set of tasks, schedule using Rate Monotonic Algorithm (RMA).
Draw the Gantt chart and determine if all deadlines are met.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2) (3)

A)

Task	C (ms)	T (ms)
T ₁	2	5
T ₂	3	8
T ₃	2	10

Schedule the tasks using the Earliest Deadline First (EDF) algorithm and prepare the Gantt chart for 0-20 ms.

Course Outcome: "CO2" / Cognitive Level: "Application"

B)

Task	C (ms)	T (ms)
T ₁	2	5
T ₂	3	8
T ₃	2	10

(3)

Using Liu & Lehoczky's criterion, find the worst-case response time for task T₃ in the above set.

Course Outcome: "CO2" / Cognitive Level: "Application"

C)

Task	C (ms)	T (ms)
T ₁	1	4
T ₂	1.5	6
T ₃	2	12

(4)

Using the analytical method of Time Demand Analysis, determine if the following task set is schedulable under Rate Monotonic Analysis (RMA).

Course Outcome: "CO2" / Cognitive Level: "Analysis"

3)

Given Periodic tasks: $\tau_1(C=1, T=5)$, $\tau_2(C=2, T=10)$

(3)

Aperiodic job: $J_1(C=3, \text{arrival}=4)$

A)

Deferrable Server: Capacity=2, Period=6

Schedule the system using EDF and Deferrable Server, showing how server budget is used and replenished.

Course Outcome: "CO3" / Cognitive Level: "Application"

B)

Given Periodic tasks: $\tau_1(C=1, T=5)$, $\tau_2(C=2, T=10)$

(3)

with an aperiodic job ($C=2, \text{arrival}=5$) which are scheduled using RMS + Slack Stealing.

Show slack intervals, response times, and CPU utilization.

Course Outcome: "CO3" / Cognitive Level: "Application"

C)

(4)

Task	C (ms)	T (ms)	Predecessors
T ₁	2	8	–
T ₂	1	12	T ₁
T ₃	2	16	T ₂

**Given three tasks with precedence constraints.
Compute the modified release times and deadlines
for scheduling under EDF**

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Explain the Priority Ceiling Protocol (PCP). (3)
Discuss how PCP avoids *deadlock* and *chained blocking* compared to PIP and HLP.

A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) (4)

Task	Priority	C (ms)	Resource Use
T ₁	High	1	R ₁ (0.5)
T ₂	Medium	2	R ₁ (1.0)
T ₃	Low	3	R ₁ (1.5)

**Illustrate priority inversion and show how Priority Inheritance Protocol (PIP) minimizes blocking.
Compute the maximum blocking time for T₁.**

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) (3)

Task	C (ms)	T (ms)
T ₁	3	8
T ₂	2	10
T ₃	4	12
T ₄	1	6

Above are Tasks for multiprocessor system.
Schedule tasks using First Fit (Bin Packing)
algorithm for two processors and compute total
utilization per processor

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Describe the Countdown Protocol and explain how it (3)

- A) ensures bounded access delays in real-time LAN communication.

Course Outcome: "CO5" / Cognitive Level: "Application"

- B) An IEEE 802.4 Token Bus network has 4 stations. (4)
Each has TTRT = 20 ms and total network utilization = 70%.
Find:

1. Maximum token holding time for each station.
2. Minimum guaranteed bandwidth

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Explain the state model of a Real-Time Operating System (RTOS) . (3)
Briefly discuss how *task states* are managed by the RTOS kernel..

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE_4412:Real time systems

#	Course Outcome	Description	Marks
1	1	CO 1	7
2	2	CO 2	13
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ELE_4412:Real time systems

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(12%)
3	Application	10	32(64%)
4	Analysis	3	12(24%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)